

Nicholas Shampoe

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SUMMARY

iOS Software Engineer specializing in media playback performance with scope expansion into Android, Web, and full-stack work. Responsible for owning the large-scale data pipeline and instrumentation behind playback decisions, driving ABR and QOE strategy.

TECHNICAL SKILLS

Programming Languages: Swift, Obj-C, C++, SQL, Java, Kotlin, TypeScript, JavaScript

iOS Frameworks: AVFoundation, UIKit, Combine, XCTest, CarPlay, FairPlay Streaming, Swift Concurrency

Tools & Platforms: AWS (Timestream, S3, CloudWatch), Git, GitHub

PROFESSIONAL EXPERIENCE

Audible (Amazon), New York

SDE 2, Playback Insights Team

March 2025 - Present

- Owned Audible's ABR algorithm end-to-end as a multi-year technical lead, including architecting a shared cross-platform code package for the core equations and mentoring teammates through the rollout. Cut on-load time-buffering by 10% across all percentiles on Android and raised the zero-buffer playback rate on Web from 96.2% to 99.8%.
- Used playback telemetry to identify serialized bottlenecks in the downloaded-playback startup path, then subsequently redesigned the flow to parallelize key processes with a single sync step. Cut load time by 31% at p50 and 32% at p99.
- Designed a system which maps internal debug error codes to user-facing, hashed error codes, giving the customer service team direct visibility into customer-reported playback issues.
- Built out QOE understanding for the playback team, distinguishing QOS metrics from actual user experience to help prioritize developer efforts on user-facing issues.
- Own the team's data pipeline (AWS Timestream, S3, CloudWatch), iterating on it over time as playback and business needs evolve. It ingests telemetry across iOS, Android, and Web, powering playback for Audible, Kindle, Amazon Music, Tesla, and Claude, the same data behind the Parallel Loading and ABR work above.

SDE 1, Playback Platform Team

June 2022 - March 2025

- Root-caused a non-regionalized legacy network call driving excessive FairPlay offline load times; then drove the fix from investigation to ship in under 30 days. Cut offline playback time-to-start by 30% overall, improving FairPlay offline p50 load time from 1.88s to 1.27s and MPEG (3p podcast) offline p50 from 0.82s to 0.45s, reaching roughly 82% of the user base.
- Traced high p99 load times to both an oversized playback manifest file and Apple's asset-processing step. Led testing and rollout of a segment-size change across the catalog that cut p90 IPD by 25% within a month of launch.
- Diagnosed a measurement bug that let backgrounded app suspensions inflate p99 load-time metrics by up to an hour, then redesigned the underlying timer logic. Improved metric accuracy by up to 27% at p99 to provide the team with a trustworthy signal for on-call alerts.
- Redesigned the team's CloudWatch metrics architecture across iOS and Android to improve accuracy without adding engineering overhead. The underlying issue hasn't recurred since, and learnings carried directly into the data pipeline iterations described above.
- Served as the team's sole Scrum Master for 18 months while mentoring teammates and interns.

AME Group Inc, Colorado Springs, Freelance App Developer

May 2021 - June 2022

- Developed "PMP Exam Prep With Aileen" iOS app to provide PMP exam candidates access to extra practice questions.
- Built and updated project requirements from kickoff through final stages, moderating customer communication and collaborating with the project owner in an Agile format.

EDUCATION

Arizona State University, Tempe, AZ

B.S. Computer Science

May 2021

M.S. Computer Science

Through 2022

Completed 27 of 30 credits before dropping the program in 2022 to join Audible.

NCAA Division 1 Athletics, Track and Field / Cross Country

August 2018 - July 2021